

Real-Time Single Channel EEG Motor Imagery based Brain Computer Interface

Jaime Camacho and Vidya Manian

Department of Electrical & Computer Engineering
University of Puerto Rico
Mayaguez, PR 00681 USA
Jaime.camacho@upr.edu, manian@ece.uprm.edu

Abstract—This paper presents a motor imagery based Brain Computer Interface (BCI) that uses single channel EEG signal from the C3 or C4 electrode placed in the motor area of the head. Time frequency analysis using Short Time Fourier Transform (STFT) is used to compute spectrogram from the EEG data. The STFT is scaled to have gray level values on which Grey Co-occurrence Matrix (GLCM) is computed. Texture descriptors such as correlation, energy, contrast, homogeneity and dissimilarity are calculated from the GLCM matrices. The texture descriptors are used to train a logistic regression classifier which is then used to classify the left and right motor imagery signals. The single-channel motor imagery classification system is tested offline with different subjects. The average offline accuracy is 87.6%. An online BCI system is implemented in openViBE with the single channel classification scheme. The stimuli presentations and feedback are implemented in Python and integrated with the openViBe BCI system.

IndexTerms—motor imagery, EEG signals, brain computer interface, single channel, time-frequency analysis, short-time Fourier transform, texture descriptors, logistic regression.

I. INTRODUCTION

Electroencephalography or EEG measures signals representing the brain's electrical activity and is acquired by placing electrodes on a subject's scalp. These electrical signals are the result of the synaptic activity of millions of neurons or nerve cells summed together. BCIs are communication pathways between the computer and a user's intent through the use of an EEG device. BCIs have been developed for motor rehabilitation, virtual reality and other gaming applications [1]. There are BCI software available for medical rehabilitation of patients with motor impediments. Most of the BCIs are driven by the subjects' intent acquired using an EEG cap and transferred to the computer to operate a device such as a computer.

Traditionally, EEG data acquisition and classification has been done utilizing any number of electrodes from 14 to 32 or 64 placed over the scalp. This results in a large amount of data and consequently entails intensive computational algorithms for processing. There are several works that report motor imagery classification utilizing spectral features extracted from the whole or subset of EEG electrodes [1-5]. Time frequency analysis of EEG signals using STFT and wavelet transform have been used for feature extraction and classification with

SVMs [6, 7]. The literature cited above present results with multiple electrodes. Single channel EEG motor imagery classification has been done on BCI datasets using STFT based time-frequency analysis in [8]. This paper presents an efficient method using single channel EEG motor imagery, time-frequency analysis, computation of time frequency texture descriptors and logistic regression to drive a BCI. Texture descriptors on EEG data have been used in [9] to detect epileptic seizures. The method shows how single channel EEG can be used for current motor imagery based BCI applications such as assisting patients with motor impediments, rehabilitation and BCI based therapy.

This paper is organized as follows. Section II presents a background on BCIs. Section III presents the methodology for the single channel EEG based BCI system. Section IV presents the complete BCI system implemented in Python and integrated with the openViBE BCI software [10]. Section V presents offline motor imagery classification results with various subjects and results of real-time online application of the method interfaced with the openViBe BCI system. Section VI presents the conclusions.

II. BRAIN COMPUTER INTERFACES

A Brain Computer Interface (BCI) aims for communication pathways between the computer or a device and the brain, based on neural activity generated by the brain. There are four main application areas of BCI systems: assisting disabled people, clinical monitoring for neurological diseases and sleeping disorders, neuroscience research on behavior, and human-machine interaction. There are some important properties [4] to take in to account in order to design a BCI system:

- noise and outliers: BCI features are noisy or contain outliers because EEG signals have a poor signal-to-noise ratio;
- high dimensionality: in BCI systems, feature vectors are often of high dimensionality. Indeed, several features are generally extracted from several channels and from several time segments before being concatenated into a single feature vector;
- time information: BCI features should contain time information as brain activity patterns are generally related to specific time variations of EEG;

- non-stationarity: BCI features are non-stationary since EEG signals may rapidly vary over time and more specifically over sessions;
- small training sets: The training sets are relatively small, since the training process is time consuming and demanding for the subjects (not have to be true for clinical use).

The basic steps for building a BCI system for human-machine interaction are 1) acquire the EEG signal 2) process and classify the EEG signal and 3) use the classified signal to control the interface 4) present a feedback to the user.



Fig. 1. Position of the ground, reference, and data acquisition node (C3 or C4).

In Figure 1, the black node is the ground, the blue node is for data acquisition and the red node is the reference node. Figure 2 shows a BCI where the EEG data acquired from the subject is processed and used to provide feedback through the display to the subject. This way the human subject and the computer interact in performing different tasks.

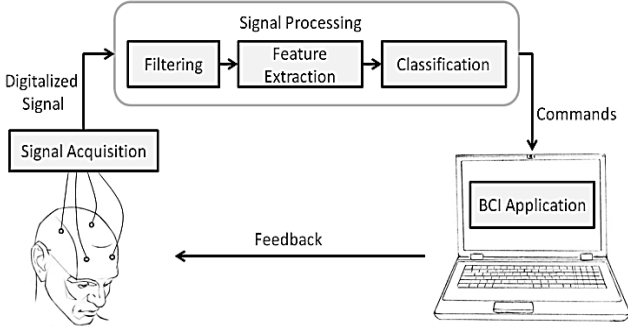


Fig. 2. A Brain Computer Interface (BCI) System.

III. SINGLE CHANNEL EEG FEATURE EXTRACTION

The steps involves EEG data acquisition from the motor area electrode C3, time-frequency analysis using the Short Time Fourier Transform (STFT), computation of the GLCM and extraction of texture descriptors, and training logistic regression classifier. Figure 3 shows the motor imagery EEG signal classification scheme. It consists of the following steps.

A. Single channel EEG data acquisition and preprocessing

The single channel EEG is obtained by placing a single electrode in the C3 area in the head. The C3 electrode is shown in Figure 1. The reference electrode and the ground are located in the motor area. The electrode selected measured the voltage on the scalp and is sampled at 400 Hz. The EEG

samples are band-pass filtered from 1-35 Hz, as these specific bands perform better in the classification of motor imagery [11].

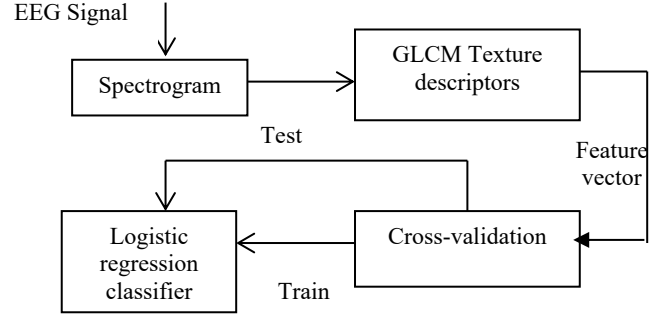


Fig. 3. Motor Imagery Feature extraction and classification scheme.

B. Time-frequency analysis using STFT

Time-frequency analyses are algorithms that allow the extraction of more information from a signal in the time-domain which has only the time and amplitude information. By applying time-frequency analysis to time domain data, the temporal and frequency characteristics can both be extracted. The new data is represented as multiple time-varying signals at different frequencies, allowing the computation of time-frequency texture descriptors from single-channel EEG signals. STFT and Wavelet transforms are the most commonly used time-frequency analysis methods.

The spectral content of the EEG signals are non-stationary, which means the signal changes over time. The Discrete Fourier transform (DFT) is a mathematical operation that decomposes a waveform into a sum of sinusoids components, where the coefficients represent the correlation between the signal and the particular frequency sinusoid. But applying the DFT, along the signal does not reveal the transitions in the spectra, it just shows the frequencies present. For that reason, applying the DFT over short periods of time (regular intervals) known as STFT is used. The EEG signal can be considered as stationary. This approach allows identification of the interval of time at which all frequencies are present in the signal. The discrete STFT is computed using a window function w centered at time n , given as:

$$I_X(n, k) = \sum_{m=-\infty}^{\infty} x[m]w[n-m]e^{\frac{-j2\pi km}{N}} \quad (1)$$

where $x[m]$ is the signal to be analyzed and N is the frequency sampling factor. The resulting STFT matrix I is represented as a matrix with time and frequency $\omega = 2\pi k/N$ information. The size of the window has the effect of changing the time-frequency resolution, with a wider window better frequency resolution but lower time resolution and vice versa for a narrow window is obtained. Figures 4 and 5 shows the spectrogram for imagining left hand and right hand movements, respectively.

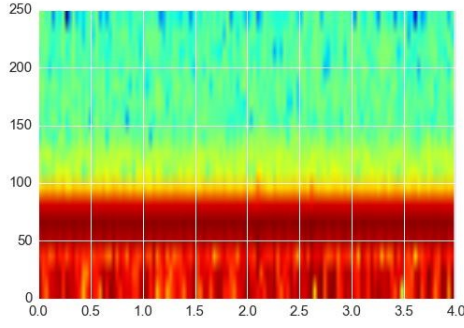


Fig. 4. Time-frequency representation of imagined right-hand movement

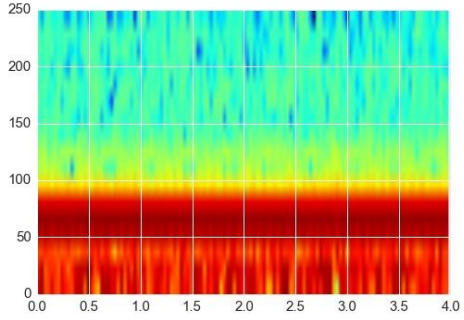


Fig. 5. Time-frequency representation of imagined left-hand movement

C. GLCM and time-frequency texture descriptors

The Gray Level Co-occurrence Matrix (GLCM) was originally proposed for gray level image classification based on texture properties. The GLCM is a second-order statistics computed on an image segment. Gray level intensities of pairs of pixels at a distance greater than or equal to 1 and at an angle of 0° , 40° , 95° or 135° degrees are compared. A matrix the size of number of gray levels is incremented at the location of each comparison. These matrices are computed for each distance and angle between pairs of pixels. They are called co-occurrence matrices. GLCM computation for an angle $\theta = 0^\circ$ and distance $\delta = 1$ is given by,

$$M_{\delta,\theta}(p, q) = \sum_{n=1}^N \sum_{m=1}^K \begin{cases} 1 & \text{if } I(n, m) = p \text{ and } I(n, m + \delta) = q \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

where $p, q = 0, 1, \dots, L-1$; L is the number of gray scales; N and K are the sizes of the STFT matrix [3]. After normalizing the GLCM, the correlation, energy, contrast, homogeneity and dissimilarity features are calculated from the GLCM matrix M . These features measure certain properties in the STFT image matrix I . Figure 6 shows the computation of these matrices. The contrast feature measures the intensity contrast between a pixel and its neighbor in the STFT image matrix I and is given by,

$$\sum_{i=1}^L \sum_{j=1}^L (i - j)^2 M_{ij} \quad (3)$$

The correlation feature measures how correlated a pixel is with its neighbor in the STFT image matrix. The range of values is 1 to -1, corresponding to perfect positive and perfect negative correlations. This measure is not defined if the

standard deviation σ_r, σ_c of the GLCM matrix is zero. The correlation feature is given by,

$$\sum_{i=1}^L \sum_{j=1}^L \frac{(i - m_r)(j - m_c) M_{ij}}{\sigma_r \sigma_c} \quad (4)$$

m_r and m_c are the mean of the rows and columns of the GLCM matrix, respectively. Energy measures the randomness of the STFT matrix and is given by,

$$- \sum_{i=1}^L \sum_{j=1}^L M_{ij} \log_2 M_{ij} \quad (5)$$

The energy is 0 when all elements of the GLCM matrix, M_{ij} are 0 and is maximum when all elements of M_{ij} are equal. The homogeneity or angular second moment measures the spatial closeness of the distribution of elements in the GLCM matrix to the diagonal. The range of values is $[0, 1]$ with the maximum being achieved when G is a diagonal matrix. It is given by,

$$\sum_{i=1}^L \sum_{j=1}^L M(i, j)^2 \quad (6)$$

Dissimilarity measures how dissimilar the elements of the GLCM matrix are, it is given by,

$$\sum_{ij} |i - j| M(i, j) \quad (7)$$

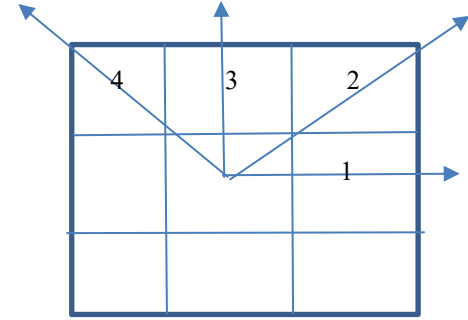


Fig. 6. Computation of GLCM.

D. Logistic regression

Logistic regression is a statistical method for analyzing a dataset in which there are one or more independent variables that determine an outcome [12]. The goal of this method is to find the best fitting model to describe the relationship between a dichotomous characteristic of interest (dependent variable; response or outcome variable) and the explanatory variable. It seeks a decision boundary to separate the given n sample-label pairs $\{(\mathbf{X}_i, y_i)\}_{i=1}^n$, where $\mathbf{X}_i \in \mathbb{R}^{st}$ is an explanatory variable in the form of a matrix and $y_i \in \{-1, +1\}$ is a dependent variable [1]. This technique transforms each feature matrix $\mathbf{X}_i$ to a feature vector $\bar{\mathbf{x}}_i = \text{vec}(\mathbf{X}_i) \in \mathbb{R}^p$, where $p = st$.

Given a new data sample $\bar{\mathbf{x}}_i$, its category can be predicted using a binomial model based on the margin $\mathbf{w}^T \bar{\mathbf{x}}_i = b$. Logistic regression construct a mapping from the feature vector $\bar{\mathbf{x}}_i$ to the label y_i . The conditional probability for classifier label y_i based on sample $\bar{\mathbf{x}}_i$, according to logistic model is

$$p(y_i | \bar{\mathbf{x}}_i, \mathbf{w}, b) = \frac{\exp[y_i(\mathbf{w}^T \bar{\mathbf{x}}_i + b)]}{1 + \exp[y_i(\mathbf{w}^T \bar{\mathbf{x}}_i + b)]}, i = 1, \dots, n \quad (6)$$

To perform the maximum likelihood estimation (MLE) of $\mathbf{w}$ and b , one can minimize the empirical loss function given by,

$$l(\mathbf{w}, b) = \frac{1}{n} \sum_{i=1}^n \log(1 + \exp[-y_i(\mathbf{w}^T \mathbf{x}_i + b)]) \quad (7)$$

IV. BRAIN COMPUTER INTERFACE SYSTEM IMPLEMENTATION

This section explains the interface between the feature extraction and classification system with the openViBE BCI for stimulus presentation and feedback. The whole system works in Python and consists of the following modules.

A. OpenViBE python module for BCI stimulus presentation

OpenViBE is an open-source software platform to design, test, and use Brain-Computer Interfaces in real and virtual environments. The purpose of OpenViBE is to obtain data from the acquisition device through the acquisition server and send it to one or more clients. The client is the OpenViBE designer. The acquisition server and the can be either on the same machine or different machines on the same network, or any combination of these [10]. The design process of a BCI using OpenViBE consists of different steps.

In the first step of the design process, a particular subject performs a specific mental task which is recorded as a training dataset. In the second step of the process, an offline analysis of the recorded training dataset is performed with the goal of finding the best calibration parameters or features for that subject. The third step consists of using the BCI application online. The process involves brain activity measurements, preprocessing, feature extraction, and classification. In the brain activity measurements, an EEG is used because the OpenViBE software supports various EEG acquisition devices. The preprocessing is important because it is where the acquired signal is denoised. In the feature extraction, few values that describe relevant information of the signal are extracted. Finally, a classifier is utilized to classify the feature vector from each class. The signal processing, which is part of the training and testing processes were implemented in Python. Also, the time-frequency analysis, and the classification process were implemented in Python, as well.

B. EEG data acquisition, feature extraction and classification modules

During the training process, the participants were seated relaxed in a chair with arms at rest. They were instructed to perform imaginary movements of left and right with their hands. Each trial started with an empty black screen displayed approximately for 10 seconds, a green cross appears to catch the participant's attention. Then, at 1 second after the green cross appeared, a red arrow was shown for 1.5 seconds pointing either to the right or to the left randomly. Each right or left position indicated by the red arrow instructed the participant to imagine a right hand or a left hand movement until the green cross disappeared for about 4 seconds of each trial. The respective imaginary movement was to last until the cross disappeared after 10 seconds. Figure 7 shows the acquisition stage. Figure 8 presents a single channel signal,

both original and filtered. The original signal is at the upper side of the Figure 8 and the filtered one is below the original signal in the same Figure. This signal was sampled at 250 Hz. It was filtered using a 4th Butterworth bandpass filter between 1-35 Hz. The filtered signal shows two high peaks; these are representations of eye blinks. Figure 9 shows the visual feedback displayed in real-time to the user for imaging left and right hand movement. This is where the training dataset of a particular subject is recorded. The classifier is trained using this dataset; specifically the data recorded using the channel C3 or C4.

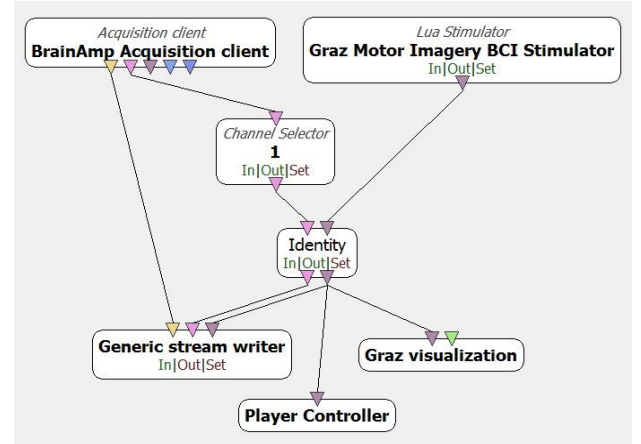


Fig. 7. Data Acquisition Stage of OpenViBE Application.

In the online testing of the experiment, after the red arrow appears, a blue rectangle appears for 1 second indicating whether the classification was left or right. The EEG signal is filtered and the STFT representation is computed. The STFT is divided into 4 parts. For each part 4 GLCM matrices are computed. For each of the matrices, 5 features are computed totaling 20 features. Hence for the 4 parts of the STFT a total of 80 features are computed. Logistic regression was trained and used for the classification process.

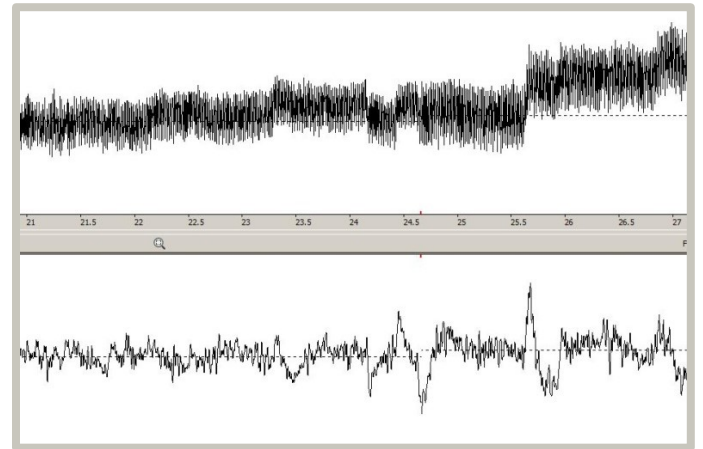


Fig 8. OpenViBE signal display for one single channel.

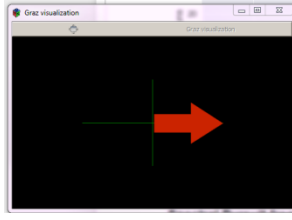


Fig. 9. OpenViBE visualization of the experiment.

V. SINGLE CHANNEL BCI SYSTEM EXPERIMENTAL RESULTS

Two sets of experiments were conducted. The first one tests offline performance of the single channel motor imagery classification scheme. 20 subjects participated in the first experiment. Each of the 20 subjects were made to sit in a comfortable chair and asked to see the monitor. On the monitor the words left and right were displayed. For each trial the subject was told to imagine moving his left and right hand based on the word in the monitor. Each subject was given several training sessions before the actual experiment. In each trial, the subjects imagines each action 11 times. Total number of trials was 120. Table I show the classification accuracies for the offline classification of motor imagery data collected from the 20 subjects. These accuracies were obtained by 10-fold cross validation.

TABLE I. CLASSIFICATION ACCURACIES FOR MOTOR IMAGERY EEG SIGNALS

Subjects	Accuracy (%)	Subjects	Accuracy (%)
Subject 1	95.62	Subject 11	77.75
Subject 2	84.96	Subject 12	77.67
Subject 3	92.60	Subject 13	99.13
Subject 4	92.40	Subject 14	95.74
Subject 5	94.67	Subject 15	87.78
Subject 6	92.79	Subject 16	98.96
Subject 7	78.36	Subject 17	81.69
Subject 8	97.02	Subject 18	79.45
Subject 9	89.26	Subject 19	80.63
Subject 10	61.31	Subject 20	94.27

Table II show the performance accuracy of the online BCI implementation using openViBE system described in Section IV.

TABLE II. Performance accuracy of the online BCI implementation.

Subject	Accuracy (%)
Subject A	97
Subject B	84
Subject C	75

Figures 10, 11 and 12 gives the Receiver Operating Characteristic (ROC) curves for new online data for Subject A, B and C using channel C4.

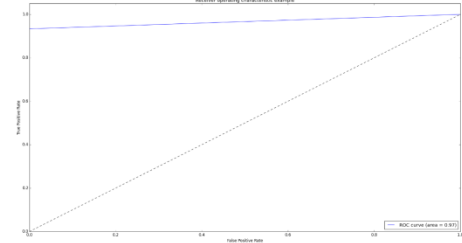


Fig. 10. ROC curve with Subject A for online BCI using channel C4.

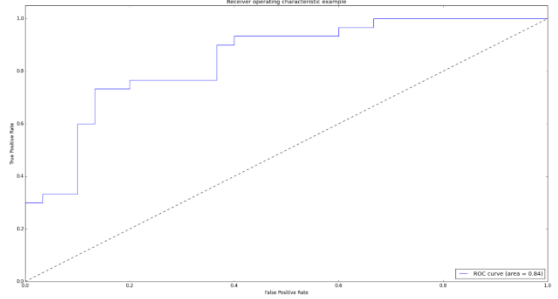


Fig. 11. ROC curve with Subject B for online BCI using channel C4.

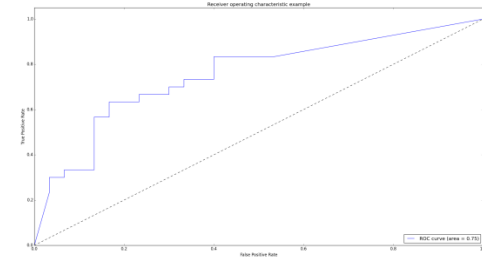


Fig. 12. ROC curve with Subject C for online BCI using channel C4.

These results show that good classification accuracies can be obtained for real-time online single channel BCI implementation by varying the parameters such as the window size of the STFT, filter order and bandwidth for each subject.

VI. CONCLUSIONS

This paper presented a real-time motor imagery based BCI that used single channel EEG signal from electrode placed in the motor area of the head, specifically C3 or C4. STFT and GLCM texture descriptors were used for time-frequency representation and feature extraction the EEG signals, respectively. Logistic regression technique was used for classification. OpenViBE BCI system was used to for stimulus presentation and feedback. The average accuracy for the offline experiment is 87.6% which shows that time-frequency texture descriptors are efficient for single channel motor

imagery classification. The accuracies obtained for offline classification in other single channel application are much lower around [8]. The advantage of single channel BCI is its ease of use in various applications such as motor rehabilitation of subjects, gaming and in development of biomedical BCI applications.

The online BCI implementation attained an average classification accuracy of 85.3% for motor imagery classification. There is a high variability between subjects A, B and C. These can be overcome by appropriately setting the parameters for each subject, as every individual has distinct EEG signals. BCI systems have to be developed keeping this fact in mind. The presented OpenViBE application can be utilized in other binary BCI interfaces. For example, it can be used in spelling and path tracing BCIs.

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